Alec Zhang

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Education

University of Maryland, College Park | **College Park, MD** December 2027
B.S. in Computer Science, Machine Learning and AI Concentration | GPA: 3.93/4.0 | Dean's List
The Harker School | San Jose, CA

Work Experience

Software Engineer September 2026 – Present
Hack4Impact UMD, SeedMoney challenge platform *College Park, MD*

- Contribute application fixes to SeedMoney's food garden challenge, which drew **660 projects in 2025**.
- Unified **completion checks** with review state so returning applicants see incomplete campaign and garden cards.

AI Engineering Intern May 2026 – August 2026
Epic Semi *Santa Clara, CA*

- Built a **3-agent** AI platform for marketing, quoting, and prospects, adopted by **one third of the company**.
- Replaced services quoted at **$2–5K/mo** with a framework-free marketing agent costing **<$10/mo in API calls**.
- Cut quoting from **about 30 steps** to **1 upload** and review-and-push, using human-reviewed Salesforce sync.
- Benchmarked **4 LLMs** in a blind bake-off, selecting Claude Sonnet after it ranked **first on all 7 ballots**.
- Deduplicated **15-minute** news polls with an **8,000-ID** ring buffer so crashed runs could resume without repeats.

Full Stack Engineer February 2026 – May 2026
Mitsubishi through AppDev *College Park, MD*

- Cut discovery **95%+**, integrating SharePoint and block-level DOCX chunking into Mitsubishi's NDA platform.
- Applied LLM edits in **under 50 ms** with a Python DOCX engine using stable block IDs and ETag concurrency.

Research Experience

MACHAMP Lab, University of Maryland January 2026 – Present
Undergraduate Researcher with Professor Jeremy Rubin *College Park, MD*

- Develop **43**-feature TEPIG group-sparse tensor regression on kidney biopsies for **1-year** post-transplant eGFR.
- Reduced mean FPR from **0.156** to about **0.0001** via adaptive thresholding; hard-cell TPR fell 0.88 to 0.18.
- Earned Excellence in Research Award for TEPIG among **550+** UMD Undergraduate Research Day participants.
- Prepare an L1000 gene-expression prediction manuscript with **real Cell Painting results** using tensor regression.

Hswen Lab, University of Maryland May 2026 – Present
Undergraduate Researcher with Professor Yulin Hswen *College Park, MD*

- Proposed and presented a **study** comparing independent LLM analyses of All of Us wearable health data with a human-specified multiverse to assess variation in conclusions.
- Designed the **LLM-agent** versus human specification-curve comparison so disagreement traces to analytic choices.

Projects

UMD Dining App | *Python, FastAPI, AWS, SwiftUI, MongoDB* December 2025 – Present

- Co-built UMD Dining for **hundreds of users** on the App Store, serving menus via a FastAPI/MongoDB API.
- Gated ranking changes on **14-day menu replays** in CI/CD, scoring junk, appeal, nDCG and personalization.
- Deployed the menu API to **AWS Elastic Beanstalk** with CI-gated auto-deploys for students using the iOS app.

Player Valuation ML Pipeline | *R, Python* 2024 – 2025

- Reached **R-squared 0.856** and **MAE about $3.82M** with a four-year model on an 80/20 split using a Python/R XGBoost pipeline testing injury history and FBref stats against Transfermarkt values.
- Reported findings in a **20-page paper**: prior market value dominates valuation; public injury history adds little.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C, Swift, Rust, OCaml, HTML/CSS, R, x86-64 Assembly
Libraries/Frameworks: React, Next.js, FastAPI, Flask, SwiftUI, SQLAlchemy, pandas, NumPy, scikit-learn, PyTorch
Tools: Git/GitHub, Docker, AWS, CI/CD, PostgreSQL, Supabase, MongoDB, Linux/macOS CLI, Azure AD/Entra ID